Etienne Rousseau

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Education

University of California Davis — Davis, CA
B.S. in Computer Science — *June 2026*

- **Courses:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks

Experience

Software Engineer — July 2026 – Present
Tesla — *Fremont, CA*

- Architected a Database-as-a-Service platform hosting Tesla's entire on-prem fleet of **10,000+** databases and VMs, reducing provisioning from **multi-day** to **under 10 minutes** and saving **hundreds of hours** per quarter
- Engineered a production **AI agent** that correlates errors across distributed trace spans via an MCP server, reducing incident response time by **over 90%** for the **100+** engineers who rely on it daily for triage
- Automated upgrades and rolling restarts across **10,000+** databases with Temporal, preventing mismanagement
- Spearheading automated configuration management for **all** database systems, enabling rapid config changes
- Designed a multi-tenant metadata catalog of databases, VMs, and clusters, unifying data for orchestration
- Developed a debugging UI consolidating database health, topology, and failures, streamlining on-call triage

Software Engineer Intern — December 2024 – June 2026
Tesla — *Fremont, CA*

- Launched a Go-based cluster identification service, slashing primary/secondary detection latency by **98.6%**
- Built a FastAPI + MongoDB metadata service with **50+** hierarchical schemas, enabling dynamic provisioning
- Implemented a topology management tool surfacing real-time metrics and failure status for **3,000+** databases
- Integrated a custom security model with upstream validation checks, enforcing service access controls

Software Engineer Intern — June 2024 – September 2024
HashiCorp — *San Francisco, CA*

- Refactored an open-source diagnostics tool into a production-grade Go library adopted across HashiCorp products
- Optimized the core processing pipeline by pruning redundant data, reducing diagnostics report latency by **27%**
- Designed extensible diagnostics outputs supporting multiple formats, improving usability for support teams
- Built a telemetry-based metrics system tracking product usage, raising platform reliability to **99.9%**
- Authored an RFC aligning multiple engineering teams on requirements, then drove the solution through to launch

Projects

Statement Analyzer | *Next.js, FastAPI, Postgres, Gemini* — April 2026 – September 2026

- Deployed a full-stack finance app that auto-categorizes bank and card transactions into **11** spending categories
- Optimized long-statement parsing by splitting text into **30K**-character chunks and skipping summary sections
- Hardened the LLM parsing pipeline against malicious content by sanitizing extracted text for prompt injection
- Integrated Plaid for live bank connections, adding transaction sync and duplicate-statement detection
- Secured every backend request with short-lived HMAC-signed tokens minted after Google OAuth sign-in

FindMyLab | *Next.js, Bun, Postgres, Docker* | findmylabdavis.org — January 2026 – June 2026

- Created a Next.js and Postgres platform helping **40,000+** UC Davis students explore **200+** research labs
- Constructed a web scraper that collects, normalizes, and maintains lab metadata from university sources
- Maintained a Bun-based backend persisting lab metadata and user-specific state across the platform
- Enabled direct student-to-lab outreach through email workflows tied to students' saved lab selections
- Added an LLM assistant that generates personalized outreach messages for contacting lab coordinators

Skills

Languages: Python, Go, Java, C/C++, C#, TypeScript, JavaScript, SQL, HTML/CSS, Bash
Developer Tools: Docker, Kubernetes, AWS, Azure DevOps, Ansible, Temporal, gRPC/Protobuf, Prometheus
Databases: PostgreSQL, MongoDB, MySQL, Redis
Frameworks: FastAPI, React, Next.js, Angular, Flask, Express, Django, Bun, Swagger